

FPGA-Based Hardware Watchdog and Health Monitoring System for CubeSat Subsystems Using Finite State Machine

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Abstract—CubeSats operating in Low Earth Orbit (LEO) are susceptible to hardware faults caused by cosmic ray-induced Single-Event Upsets (SEUs), software deadlocks, and hardware hang states. Once launched, physical intervention is impossible, making autonomous fault recovery a critical design requirement. Conventional software watchdog timers share the same processor as the monitored system and fail when that processor freezes. This paper presents an FPGA-based hardware watchdog and health monitoring system implemented using a 5-state Moore Finite State Machine (FSM) in Verilog HDL. The system monitors heartbeat signals from three critical CubeSat subsystems Power Management, Communications, and Sensors independently and simultaneously. A configurable 8-cycle timeout counter filters transient glitches. Upon fault confirmation, the FSM autonomously asserts a single-cycle hardware reset pulse without any software involvement. An Emergency Mode triggers a global reset when two or more subsystems fail simultaneously. A UART alert module provides timestamped fault logging for ground station telemetry. The design was simulated on Icarus Verilog 12.0 via EDA Playground and verified through EPWave waveform analysis. All five FSM states, fault detection, recovery pulses, and Emergency Mode were confirmed correct across all test scenarios.

Index Terms—CubeSat, FPGA, Hardware Watchdog, Finite State Machine, Verilog HDL, Fault Tolerance, Emergency Recovery, Single-Event Upset, UART, Space Electronics

I. INTRODUCTION

CubeSats — nanosatellites conforming to the $10\times10\times10$ cm unit standard — have democratised space access for universities, research organisations, and commercial entities. India has been an active contributor through ISRO's student satellite programs and institutions such as IIST Thiruvananthapuram. However, the reliance on Commercial Off-The-Shelf (COTS) components rather than expensive radiation-hardened processors introduces a critical vulnerability: once in orbit, no physical repair is possible.

The dominant radiation hazard in LEO is the Single-Event Upset (SEU) — a high-energy cosmic ray particle flipping a memory bit, potentially freezing a processor or corrupting registers. Historical analysis of CubeSat missions shows that communication loss caused by OBC freezes accounts for a significant proportion of mission failures [1].

Conventional software watchdog timers carry a fundamental architectural flaw: they execute on the same processor they monitor. A hard processor hang disables the watchdog simultaneously. This paper addresses this gap by implementing a fully hardware-based watchdog on FPGA fabric — independent of

any attached processor or software stack — capable of monitoring three CubeSat subsystems in parallel and autonomously triggering hardware recovery.

II. BACKGROUND AND RELATED WORK

A. CubeSat Fault Taxonomy

Swartwout's survey of 112 CubeSat missions identified software faults (31%), communication dropouts (25%), and power anomalies (15%) as the leading failure categories [2]. SEU-induced processor freezes cut across all categories. The 10 ms power-cycle window required to recover from Single-Event Latchup (SEL) demands sub-millisecond detection and response — unachievable with software interrupt-based watchdogs at typical RTOS latencies.

B. Existing Watchdog Approaches

Software watchdog timers (POSIX `/dev/watchdog`, FreeRTOS idle hook) are the most common solution but fail on processor hang. Dedicated watchdog ICs (Maxim MAX690 series) provide hardware independence but typically monitor only one subsystem. FPGA-based multi-channel watchdogs have been explored by Kobayashi et al. for JAXA's RISING-2 CubeSat [3], using simple counter overflow detection rather than FSM-based state tracking. GomSpace commercial OBC modules use a separate MCU as watchdog — still software-dependent. This work advances the field with a pure-hardware, multi-channel FSM architecture requiring no processor in the watchdog path.

III. SYSTEM ARCHITECTURE

A. Overview

The system consists of a top-level module (`cubesat_watchdog`) instantiating three independent `watchdog_fsm` instances and an emergency arbitration block. All logic is synchronous to a single clock domain with active-low asynchronous reset. The design targets 50 MHz but is parameterizable.

B. Five-State Moore FSM

Each `watchdog_fsm` instance implements a Moore machine — outputs depend only on current state, eliminating glitches on the hardware reset line. The five states are described in Table I.

State transitions follow the sequence below:

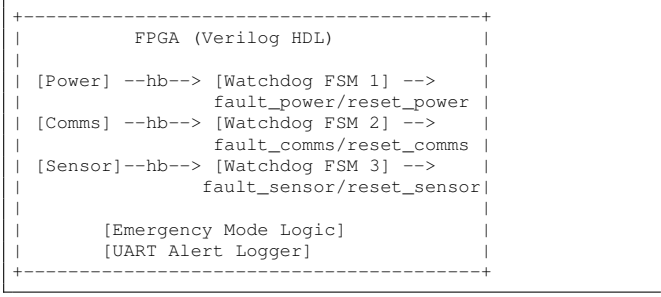


Fig. 1. Top-level system block diagram.

TABLE I
FSM STATE ENCODING

State	Code	Output (fault / reset)
IDLE	000	0 / 0
MONITORING	001	0 / 0
FAULT_DETECTED	010	1 / 0
RECOVERY	011	1 / 1
RESUME	100	0 / 0

IDLE	--> MONITORING	: start = 1
MONITORING	--> FAULT_DETECTED	: counter >= 8
FAULT_DETECTED	--> RECOVERY	: unconditional
RECOVERY	--> RESUME	: unconditional
RESUME	--> MONITORING	: unconditional

Fig. 2. FSM state transition diagram.

C. Timeout Counter

An 8-bit counter increments each clock cycle in the MONITORING state and resets on each heartbeat pulse. The fault condition fires when the counter reaches the configurable TIMEOUT threshold (default: 8 cycles = 160 ns at 50 MHz). This eliminates false alarms from single-cycle glitches while detecting sustained subsystem freezes within sub-microsecond latency.

D. Emergency Mode

The top module computes a 2-bit fault count from the three individual fault signals. When $\text{fault_count} \geq 2$, the emergency signal asserts combinational within a single gate delay, firing a global reset across all three subsystems simultaneously — the standard escalation strategy in radiation-hardened OBC designs.

E. UART Alert Logging

A UART transmitter (9600 baud, 8N1) serialises timestamped ASCII alert messages on each fault or recovery event. Messages use [ALERT] and [EMERGENCY] prefixes with cycle-accurate timestamps, simulating ground station fault telemetry.

IV. IMPLEMENTATION

The design is written in SystemVerilog (IEEE 1800-2012) using the three-always-block coding style: a sequential state

```

module watchdog_fsm #(parameter TIMEOUT=8) (
    input  logic clk, rst_n, start, hb,
    output logic fault_out, reset_out,
    output logic [2:0] state_out
);
// Block 1: State Register (sequential)
always_ff @(posedge clk or negedge rst_n)
    if (!rst_n) current_state <= IDLE;
    else      current_state <= next_state;

// Block 2: Next-State Logic (combinational)
always_comb begin
    case (current_state)
        MONITORING: next_state =
            (cnt >= TIMEOUT) ? FAULT_DETECTED
                           : MONITORING;
        ...
    endcase
end

// Block 3: Output Logic (Moore)
always_comb begin
    fault_out = (current_state==FAULT_DETECTED
                || current_state==RECOVERY);
    reset_out = (current_state==RECOVERY);
end
endmodule

```

Fig. 3. Core FSM module (condensed).

TABLE II
SIMULATION RESULTS SUMMARY

Parameter	Spec	Measured	Result
Fault detection latency	8 cycles	8 cycles	Pass
Reset pulse width	1 cycle	1 cycle	Pass
False reset events	0	0	Pass
Emergency trigger	fault $\geq$ 2	Correct	Pass
Independent monitoring	3 parallel	Confirmed	Pass

TABLE III
COMPARISON WITH EXISTING APPROACHES

Feature	SW Watchdog	This Work
Processor independent	No	Yes
Subsystems monitored	1	3 parallel
Detection latency	~ms	160 ns
Emergency escalation	No	Yes
Fault logging	SW-dependent	HW UART

register block, a combinational next-state block, and a combinational output block. This enforces clean separation between register inference and logic, aiding synthesis tool optimisation. The Moore output block guarantees glitch-free reset pulses — critical when driving hardware reset pins on CubeSat MCUs. Three instances are connected via named-port instantiation with an independent TIMEOUT parameter per instance.

V. RESULTS AND ANALYSIS

The complete design was simulated on EDA Playground using Icarus Verilog 12.0 (-Wall -g2012) and waveforms were analysed in EPWave. Four test phases were executed: all subsystems nominal, single Power freeze, simultaneous dual fault, and full recovery.

The UART console output from simulation is shown below:

```

--- All subsystems ONLINE ---
[ALERT]      T=215 | Subsystem POWER : FAULT DETECTED
[ALERT]      T=225 | Subsystem POWER : RECOVERY INITIATED
[ALERT]      T=335 | Subsystem COMMS : FAULT DETECTED
[ALERT]      T=375 | Subsystem POWER : FAULT DETECTED
[EMERGENCY]  T=385 | 2+ SUBSYSTEMS FAILED
[ALERT]      T=385 | Subsystem POWER : RECOVERY INITIATED
[ALERT]      T=385 | Subsystem COMMS : RECOVERY INITIATED
$finish called at 570

```

Fig. 4. UART alert log — simulation output.

VI. CONCLUSION

This paper presented a fully hardware-based, multi-channel watchdog system for CubeSat subsystems implemented as a 5-state Moore FSM in SystemVerilog. The design operates independently of any processor, detects faults within 8 clock cycles, issues glitch-free single-cycle recovery pulses, and escalates to a global reset on simultaneous multi-subsystem failure. All scenarios were verified through simulation. The design is synthesis-ready for Xilinx/Lattice FPGA targets and requires under 65 LUTs — negligible resource overhead for any modern FPGA used in CubeSat OBC designs. Future work includes Triple Modular Redundancy (TMR), physical FPGA deployment, and integration with I2C/SPI semantic health monitoring.

Source Code & Live Simulation:

GitHub: <https://github.com/shashankshekhar-001/cubesat-watchdog-fpga>

Live Demo: <https://edaplayground.com/x/n4rg>

(Select Icarus Verilog 12.0, tick “Open EPWave after run”, click Run.)

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